

	GLOBAL ECONOMICS COMPUTATIONAL THINKING GRADE: 11TH TERM 2 – PROJECT EXCEL t-STUDENT CONFIDENCE INTERVAL PROJECT Nicolás López Cuéllar, Jaime David Navarrete	SECOND TERM 2025–2026
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Learning objective: Present and defend an Excel tool that estimates a confidence interval for the mean with unknown variance (t-student), using dashboards and simple macros to support contextual conclusions in finance and economy.	Assessment criteria: Global Economics C10: Concludes about a statistical parameter in situations in finance and economy. Computational Thinking C8: Infers conclusions based on data analysis results. C9: Justifies conclusions through interactive dashboards. C10: Formulates solutions to different scenarios described by data using dashboards and simple macros.
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1. Project brief

1.1 Purpose of the project

This interdisciplinary project requires students to design and implement an Excel workbook that receives real economic or financial data and estimates confidence intervals for the population mean when variance is unknown. The workbook must make the procedure simple, transparent, and repeatable through dashboards and short macros. The presentation and defense of this project are the direct evidence for C10 in both subjects. Presenting this project will also allow the student to present problems for:

- C8: Calculates confidence intervals applying t-student distribution in context situations.
- C9: Estimates the confidence interval for the mean when the variance is unknown in context situations.

1.2 Computational Thinking Background

Before this project, students have already worked with:

- C1: Integrates external data sources into Excel.
- C2: Understands array formulas in spreadsheets.
- C3: Uses array formulas in spreadsheets.
- C4: Understands macros in spreadsheets.
- C5: Uses macros to automate repetitive tasks in spreadsheets.
- C6: Identifies Data Analysis tools in spreadsheets.
- C7: Uses Data Analysis tools in spreadsheets.

1.3 Input specification

The Excel workbook must receive:

- A dataset of numerical observations (raw data), typed, pasted, or imported from an external source.
- Optional metadata: context label, units, date range, and scenario label.
- A confidence level selector (for example 90%, 95%, or 99%).

1.4 Processing requirements

The workbook must:

- Compute sample size n , sample mean $\bar{x}$, sample standard deviation s , and standard error $SE = \frac{s}{\sqrt{n}}$.
- For unknown variance, compute and report t-based outputs: t statistic/critical value used for the interval (t^* or equivalent), margin of error $ME_t = t^* \cdot SE$, and confidence interval $CI_t = (\bar{x} - ME_t, \bar{x} + ME_t)$.
- Compute and report z-based approximation outputs: z statistic/critical value used for the interval (z^* or equivalent), margin of error $ME_z = z^* \cdot SE$, and confidence interval $CI_z = (\bar{x} - ME_z, \bar{x} + ME_z)$.
- Provide a clear conclusion template in context language, prioritizing the t-based interval when variance is unknown.
- Use dashboards (dropdowns, slicers, scenario selectors, and charts) to justify conclusions.

- Use simple macros to:
 - clear or reset inputs,
 - recompute and snapshot outputs for different scenarios,
 - export or log scenario results in a comparison table that stores both CI_t and CI_z .

1.5 Output specification

For each dataset and selected confidence level, the workbook output must include:

- **t-based outputs (variance unknown):** n , $\bar{x}$, s , $SE = \frac{s}{\sqrt{n}}$, t^* (or equivalent t statistic), $ME_t = t^* \cdot SE$, and $CI_t = (\bar{x} - ME_t, \bar{x} + ME_t)$.
- **z-based approximation outputs:** z^* (or equivalent z statistic), $ME_z = z^* \cdot SE$, and $CI_z = (\bar{x} - ME_z, \bar{x} + ME_z)$.
- Dashboard visuals supporting the justification.
- A scenario comparison view with at least two scenarios, supported by dashboard and/or macro outputs.

1.6 Comparison requirement

The dashboard must display CI_t and CI_z side-by-side and include a short interpretation comparing interval width differences, when both methods are close, and the role of sample size in reducing the t-vs-z gap.

1.7 Dashboard and workflow requirement

1. Import or paste raw data and define meta-data and confidence level.
2. Execute workbook calculations to obtain both t-based and z-based outputs.
3. Use a dashboard toggle or side-by-side view for t vs z values, including explicit visualization of both intervals.
4. Report $ME_t - ME_z$ and/or interval-width difference, and use scenario selectors to compare contexts and justify conclusions.

2. Solved Example 1

Financial context: daily exchange-rate spread in a broker desk

A brokerage team analyzes the daily spread (basis points) observed in a currency operation window. Raw data are entered as individual observations in Excel:

Spread data (bp)
49.2, 51.4, 47.8, 52.1, 50.5, 48.9, 53.0, 49.7, 50.9, 51.8, 48.6, 52.4

The dashboard selector is set to **95% confidence** and scenario **Base week**.

Workbook output

Metric	Value
n	12
$\bar{x}$ (bp)	50.52
s (bp)	1.68
SE	0.484
t^* (95%, $df = 11$)	2.201
ME_t	1.06
CI_t lower bound	49.46
CI_t upper bound	51.59
z^* (95% normal approx.)	1.960
ME_z	0.95
CI_z lower bound	49.57
CI_z upper bound	51.47
$ME_t - ME_z$	0.11

Conclusion

With 95% confidence, using the preferred t-based method for unknown variance, the mean daily spread for the selected week is between 49.46 and 51.59 basis points. The z-approximation interval is 49.57 to 51.47 basis points, which is close but slightly narrower.

3. Solved Example 2

Economic context: import shipping cost per container

An import company records shipping cost per container (thousand USD) for one route. Data are imported from CSV to Excel as individual observations:

Cost data (thousand USD)
1.72, 1.81, 1.75, 1.88, 1.93, 1.79, 1.85, 1.90, 1.77, 1.84, 1.96, 1.87, 1.80, 1.92, 1.74, 1.89, 1.83, 1.95

The dashboard selector is set to **90% confidence** and the route scenario **Port A – Q2**.

Workbook output

Metric	Value
n	18
$\bar{x}$ (thousand USD)	1.844
s (thousand USD)	0.074
SE	0.017
t^* (90%, $df = 17$)	1.740
ME_t	0.030
CI_t lower bound	1.814
CI_t upper bound	1.874
z^* (90% normal approx.)	1.645
ME_z	0.028
CI_z lower bound	1.816
CI_z upper bound	1.872
$ME_t - ME_z$	0.002

Conclusion

With 90% confidence, using the preferred t-based method for unknown variance, the mean shipping cost parameter for this route is between 1.814 and 1.874 thousand USD per container. The z-approximation interval is 1.816 to 1.872 thousand USD, very close to the t-based result in this case.